Question 1

Find which Linux kernel version is currently running.

Approach

Use uname command to display kernel release information.

Commands

- `uname -r` → shows kernel version
- `uname -a` → shows full system kernel details

Output

STEP 1 : display kernel version

```
┌──(aman㉿kali)-[~]
└─$ uname -r
6.18.12+kali-amd64
```

Key Observations

Kernel version identifies system capabilities.
Important for exploit compatibility checks.

What I learned

Kernel version helps determine vulnerability targets.

Question 2

Show all currently loaded kernel modules.

Approach

Use lsmod to list active kernel modules in memory.

Commands

- `lsmod` → lists loaded kernel modules

- `less` → scroll output

Output

STEP 1 : list kernel modules

```
┌──(aman㉿kali)-[~]
└─$ lsmod
Module                  Size  Used by
snd_seq_dummy          12288  0
snd_hrtimer            12288  1
snd_seq               114688  7 snd_seq_dummy
snd_seq_device         16384  1 snd_seq
qrtr                   57344  2
rfkill                 45056  2
vboxsf                 49152  0
sunrpc                868352  1
binfmt_misc            28672  1
snd_intel8x0           57344  1
snd_ac97_codec        208896  1 snd_intel8x0
ac97_bus               12288  1 snd_ac97_codec
intel_rapl_msr         20480  0
intel_rapl_common      53248  1 intel_rapl_msr
snd_pcm               212992  2 snd_intel8x0,snd_ac97_codec
snd_timer              49152  3 snd_seq,snd_hrtimer,snd_pcm
snd                   159744  10 snd_seq,snd_seq_device,snd_intel8x0,snd_timer,snd_ac97_codec,snd_pcm
soundcore              16384  1 snd
joydev                 32768  0
vboxguest              53248  7 vboxsf
pcspkr                 12288  0
ac                     16384  0
sg                     45056  0
evdev                  32768  10
configfs               61440  1
efi_pstore             12288  0
nfnetlink              20480  1
vsock_loopback         12288  0
vmw_vsock_virtio_transport_common    57344  1 vsock_loopback
vmw_vsock_vmci_transport    49152  0
vsock                  69632  3 vmw_vsock_virtio_transport_common,vsock_loopback,vmw_vsock_vmci_transport
vmw_vmci              102400  1 vmw_vsock_vmci_transport
autofs4                61440  2
ext4                 1224704  1
crc16                  12288  1 ext4
mbcache                20480  1 ext4
jbd2                  196608  1 ext4
crc32c_cryptoapi       12288  0
hid_generic            12288  0
usbhid                 86016  0
hid                   274432  2 usbhid,hid_generic
vmwgfx                487424  2
drm_ttm_helper         16384  2 vmwgfx
sd_mod                 86016  2
ttm                   131072  2 vmwgfx,drm_ttm_helper
sr_mod                 28672  0
cdrom                  81920  1 sr_mod
drm_client_lib         12288  1 vmwgfx
```

Key Observations

Shows module name, size, dependencies.
Loaded modules increase attack surface.

What I learned

Kernel modules extend OS functionality dynamically.

Question 3

Turn on packet forwarding using sysctl command.

Approach

Modify kernel runtime parameter to allow system to forward network packets.

Commands

- `sysctl` → modify kernel parameters
- `net.ipv4.ip_forward` → controls packet forwarding

Output

STEP 1 : enable IP forwarding

```
┌──(aman㉿kali)-[~]
└─$ sudo sysctl -w net.ipv4.ip_forward=1
[sudo] password for aman:
net.ipv4.ip_forward = 1
```

Key Observations

Change applies immediately in memory.
Used for routing and MITM setups.

What I learned

Kernel parameters can be changed without reboot.

Question 4

Modify sysctl configuration file to control packet forwarding permanently, then disable it.

Approach

Edit persistent configuration file and reload settings.

Commands

- `nano` → edit configuration file
- `sysctl -p` → apply config changes
- `net.ipv4.ip_forward=0` → disable forwarding

Output

STEP 1 : edit sysctl configuration

```
┌──(aman㉿kali)-[~]
└─$ sudo nano /etc/sysctl.conf
```

STEP 2 : disable IP forwarding

```
┌──(aman㉿kali)-[~]
└─$ sudo sysctl -w net.ipv4.ip_forward=0
net.ipv4.ip_forward = 0
```

STEP 3 : apply configuration

```
┌──(aman㉿kali)-[~]
└─$ sudo sysctl -p
net.ipv4.ip_forward = 0
```

Key Observations

Persistent settings survive reboot.
Misconfiguration may expose routing capability.

What I learned

System-wide kernel behaviour can be permanently configured.

Question 5

Inspect detailed information about a specific kernel module.

Approach

Use modinfo to retrieve metadata about module such as author and dependencies.

Commands

- `modinfo` → shows module details

Output

STEP 1 : get module information

```
┌──(aman㉿kali)-[~]
└─$ modinfo ext4
filename:       /lib/modules/6.18.12+kali-amd64/kernel/fs/ext4/ext4.ko.xz
license:        GPL
description:    Fourth Extended Filesystem
author:         Remy Card, Stephen Tweedie, Andrew Morton, Andreas Dilger, Theodore Ts'o and others
alias:          fs-ext4
alias:          ext3
alias:          fs-ext3
alias:          ext2
alias:          fs-ext2
depends:        jbd2,crc16,mbcache
intree:         Y
name:           ext4
retpoline:      Y
vermagic:       6.18.12+kali-amd64 SMP preempt mod_unload
sig_id:         PKCS#7
signer:         Build time autogenerated kernel key
sig_key:        6A:3D:16:CE:5B:74:79:47:D7:15:22:11:41:56:07:99:34:D9:38:2F
sig_hashalgo:   sha256
signature:      30:66:02:31:00:86:01:2E:5B:47:62:33:45:D8:4D:DA:8D:AF:E6:D3:
                A2:81:02:CB:A2:C6:03:06:B2:57:1F:CB:CF:F9:C2:49:8F:A9:76:62:
                46:FE:BF:97:81:88:71:4F:2A:78:F5:32:3D:02:31:00:89:3F:57:2E:
                7A:45:78:FD:77:F5:AE:6C:B6:86:79:E1:1F:D0:BF:7F:90:34:4F:5D:
                96:96:EB:4C:DC:FE:C9:E2:12:29:C7:29:E4:B9:C9:56:42:FD:3F:02:
                22:23:41:C6
```

Key Observations

Shows module parameters and description.
Useful for understanding kernel features.

What I learned

kernel modules include metadata helpful for analysis.

Security Insights

1. Kernel version helps attackers identify exploitable vulnerabilities.
2. Loaded modules reveal potential privilege escalation vectors.
3. IP forwarding enables traffic interception attacks.
4. Persistent kernel configs can create long-term backdoors.
5. Module metadata helps attackers understand system internals.